Building a Virtual Knowledge Graph for Data Lakes Using Croissant Metadata

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Abstract

Virtual Knowledge Graphs (VKGs) expose heterogeneous data sources through a single SPARQL interface, relying on mappings between an ontology and the underlying sources. In data lakes, where new data sources are constantly added, building these mappings by hand does not scale. Further, data lakes also store semi-structured data sources, like CSV files, where schema information is also often missing or opaque. In these cases, automatic mapping generation, also called bootstrapping, is the most pragmatic choice. In this paper, we propose addressing this challenge using Croissant, a metadata vocabulary to describe datasets. We show how using such annotations allows to derive both a shallow ontology and R2RML mappings that enable effective VKG integration. This strategy allows us to build the data management layer within the DataGEMS platform, an EU funded dataset discovery and management system. We implement our pipeline using the Ontop Virtual Knowledge Graph system and the data federator Dremio. We present early results on CSV files and relational data, as well as identified technical issues and interesting research challenges within the proposed setup.

Keywords

R2RML Mappings, Croissant Metadata, Virtual Knowledge Graphs, Data Lakes

1. Introduction

Data lakes serve as state-of-the-art flexible systems for collecting and storing heterogeneous datasets from multiple sources in large organizations [1, 2]. A data lake can often store relational tables along with semi-structured data, like CSV files, other tabular formats like parquet files, and XML or JSON documents. In some cases, they also store textual documents along with images and videos. The advantage offered by data lakes is to avoid upfront integration efforts: sources can be stored in their own format, without the requirement to adhere to a specific schema. However, these sources remain interesting to query, and particularly to be queried together. Yet, given the datasets stored in a data lake are heterogeneous, there is no obvious common query language that allows cross-source querying.

Virtual Knowledge Graph (VKG) systems [3, 4] implement the Ontology-Based Data Access (OBDA) paradigm to provide a unified query interface, using the SPARQL language, to various structured data sources. Such systems expose databases as virtual RDF graphs by linking ontology terms (classes and properties) to data sources via mappings. These mappings and the ontology are inputs to the system that work with a Global-as-View (GAV) approach. Such mappings can be described using the W3C standard R2RML [5], or native mapping language such as obda mappings for Ontop [4].

Generating mappings is a central challenge in Virtual Knowledge Graph construction because it requires aligning heterogeneous data structures with ontology terms in a consistent and semantically meaningful way. Two main families of approaches have emerged: bootstrapping approaches [6, 7, 8, 9] and ontology-driven mapping generation approaches [10, 11]. Bootstrapping approaches [6, 7, 8, 9] automatically derive both an ontology and mappings from the structure of a relational source, typically following W3C Direct Mapping [12] recommendation. Systems such as BootOX and MIRROR enrich this process by inferring implicit information that is normally contained in relational databases (e.g., subclass-of relationships, M-N relationships), and provenance information. In contrast, ontology-driven mapping generation approaches [10, 11] assume that a domain ontology already exists and focus on

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automatically producing mappings that connect source attributes to ontology classes and properties. Such approaches leverage schema information to establish correspondences between the ontology and the data source, and use these correspondences to generate the mappings. In both families, the mapping generation focuses on relational databases, whereas R2RML mappings can support any data source that can be queried with SQL, e.g., CSV files. Further, existing approaches primarily rely on relational schema information [7, 6, 10, 11], while datasets stored in data lakes often have limited or inconsistent metadata, making automatic extraction with these approaches more challenging.

Recently, Croissant [13] has been proposed as a metadata schema and vocabulary extension for machine learning datasets. A Croissant [13] document is a JSON-LD document describing a dataset and the raw files it contains, how they are structured, and how they connect. Therefore, a Croissant document captures both structural and semantic information beyond the schema. This allows popular ML frameworks to ingest the dataset directly with minimal code.

Contributions. In this work, we demonstrate how Croissant metadata provides an opportunity to enable automatic mapping generation by exploiting the additional semantic context available in dataset profiles. Therefore, in this work, we introduce an approach for automatically generating an ontology and the R2RML mappings from Croissant [13] profiles of data lake sources. We also introduce the integration pipeline of the data sources into the virtual knowledge graph. This approach is being deployed within the DataGEMS [1] data discovery platform, a European Commission funded research project aiming to provide a new data management platform for data scientists. The DataGEMS system continuously ingests heterogeneous datasets originating from diverse sources, many of which lack rich or reliable schema information. The platform adopts Croissant [13] as its internal metadata framework and automatically generates Croissant profiles to describe dataset structure and semantics. Thanks to this, the platform aims to integrate new datasets into a Virtual Knowledge Graph, exploiting their Croissant dataset profiles.

Exemplary Use Case. The DataGEMS platform includes, among others, meteorological data stored in different formats and systems. Station metadata are provided through a CSV file, while historical observations are maintained in a PostgreSQL database. The integration of these sources allows users to query meteorological observations together with the corresponding station information without needing to access each data source separately.

2. Integrating Data Sources into a Virtual Knowledge Graph System

We propose a pipeline to integrate data sources into a Virtual Knowledge Graph by bootstrapping VKG mapping from Croissant dataset descriptors (Figure 1). The proposed pipeline includes a Virtual Knowledge Graph system and a data federator that can communicate between the data sources and the Virtual Knowledge Graph to support heterogeneous data sources. During the integration of a new data source, a Croissant profile is generated to provide a standardized metadata description while the source itself is registered into the data federator. Given the croissant profile, both the ontology and the mappings are generated and added to the Virtual Knowledge Graph system.

2.1. Croissant Profile for Dataset Metadata Representation

Dataset metadata is essential for dataset discovery, reproducibility, and automated processing, but existing ecosystems often rely on heterogeneous metadata formats, making interoperability difficult. The Croissant specification [13] addresses this challenge by providing a unified vocabulary to describe datasets and their associated resources. Croissant [13] is an extension of the schema.org [14] vocabulary, introducing a set of nodes and properties to represent dataset metadata. A Croissant profile provides general information about a dataset, including its name, description, license, access URL, and other schema.org common attributes, while also describing its internal structure and data organization. It

[1] https://datagems.eu/

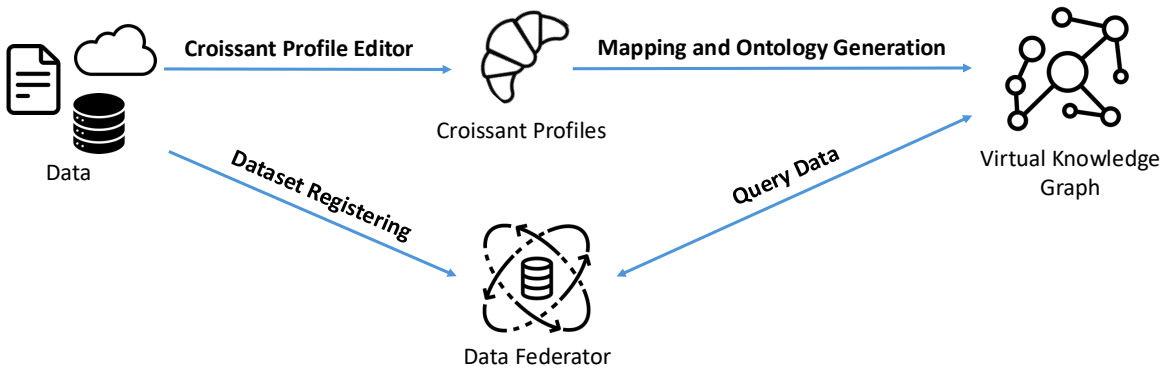


Figure 1: Integration of Data Lake Sources into a Virtual Knowledge Graph

represents the contents of a dataset through `FileObject` and `FileSet` nodes, and describes file contents via `RecordSets` and their associated *Fields*. These fields can represent simple attributes, such as table columns for relational tables or CSV files, or more complex structures for other data types. The Croissant framework does not require to modify the original data format, instead it introduces a metadata layer that exposes a unified view over the data. For this reason, it is also possible to automatically generate croissant profiles describing the content of a given archive, e.g., with tools such as Croissant Baker [15].

2.2. Mapping and Ontology Generation

Given a Croissant profile, i.e., a JSON-LD representation of croissant metadata describing a given archive, our approach generates a simple ontology and direct mappings for each file described in the archive. The ontology relies on the dataset record sets: each `RecordSet` of the dataset is translated into an ontology class, and each `Field` of this `RecordSet` is a property. The provenance, i.e., the IRI of `Field` and `RecordSet` nodes, is added to the generated ontology. The Croissant vocabulary further allows the expression of primary and foreign keys. Foreign keys are used to generate object properties, while Field nodes without any foreign keys are considered datatype properties.

The approach follows the Direct Mapping recommendations to generate mappings. For relational databases, these mappings also extend to include implicit schema information, such as many-to-many tables, as proposed in BootOX [6]. When provided, the primary key is used to generate the template-valued term map, i.e., the IRI value for a specific row. However, in cases like CSV files, such a primary key is rarely defined and cannot be trivially derived. This poses the first important challenge in this content. To overcome this challenge, we propose to generate a Blank Node for each row. Blank node identifiers generated this way are not reusable across queries, e.g., the same instance retrieved in two separate SPARQL queries will not share an identifier, which prevents, for example, caching a weather station's IRI from one query to reuse it in a second query.

Figure 2 shows part of a dataset croissant description and all the generated files: ontology, mappings and lenses. The proposed example contains a primary key, the field *id*, that is used to generate the IRI template of a weather variable. If no primary key is defined, then instead of the current template, a Blank Node will be generated.

2.3. Experimental Setup

This approach has been implemented[2] within the DataGEMS project. We selected Ontop [4] as the Virtual Knowledge Graph system since it is a well-maintained and open-source system, and Dremio [16] as the data federator. Other federators, such as Denodo [17] or Teiid [18], are also compatible with Ontop. We are currently validating this approach integrating both relational databases (stored in

[2]https://github.com/VIASystemsLab/DataLake-VKG

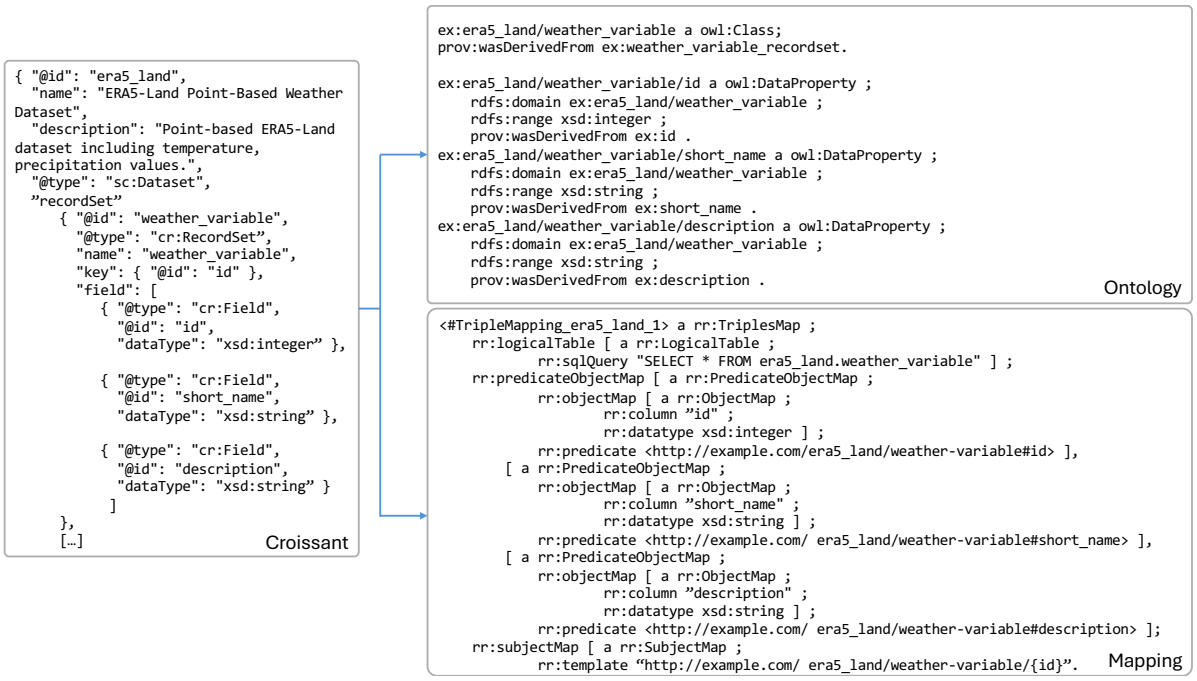


Figure 2: Mapping and Ontology Generation Example from Croissant

```
{
  "relations": [
    {
      "name": ["lenses", ”era5_land”, ”weather_variable”],
      "baseRelation": [”era5_land", ”weather_variable"],
      "uniqueConstraints": {
        "added": [
          { "name": ”era5_land_weather_variable", "determinants": [”id"] }
        ]
      },
      "type": "BasicLens"
    }
]}
```

Lenses

Figure 3: Example of Lenses Generated from Croissant

PostgreSQL) and CSV files stored in a S3 compatible object storage. The data integration workflow is accessible via API: anytime a dataset is added to the platform, a croissant profile is automatically generated and sent to the API endpoint to be registered. Our system, then, registers the data source in Dremio, and generates the mapping and ontology files for this data source.

We found that, for Ontop, generating the R2RML mapping alone does not provide to the system enough information to produce efficient queries. Even when primary keys are defined in the data source, in the configuration with Dremio, Ontop-generated queries perform multiple unnecessary self-joins. To ensure primary keys are correctly handled instead, Ontop supports the use of *lenses files*[3]. Our approach is able to also generate lenses files using the Croissant profile. Nonetheless, if no primary-key information is provided, lenses cannot be of help, hence large data sources could become unqueryable due to unnecessary self-joins in the generated queries making them computationally unfeasible.

Figure 3 shows an example of the generated lenses file for example introduced in Figure 2. The mappings should include the location of the dataset through the corresponding lenses file. Concretely, the value of `name` defined in the lenses files (e.g., `"lenses"."era5_land"."weather_variable"`) is referenced in the *sqlQuery* section of the mapping file. If no relation is defined for a specific part, i.e, if no primary or foreign key is defined, then the *sqlQuery* section of the mapping file should only contain the location of the data source.

[3]https://ontop-vkg.org/guide/advanced/lenses.html

Table 1
Performance Comparison With and Without a Lenses File

		Join Query	Postgres Query	CSV Query
Query Execution	With Lenses	4.87 ± 0.40	0.59 ± 0.11	4.69 ± 0.47
	Without Lenses	829.88 ± 743.28 (*)	349.03 ± 3.33 (*)	8.68 ± 1.23
Ontop Rewriting	With Lenses	0.007 ± 0.15	0.01 ± 0.02	0.01 ± 0.01
	Without Lenses	0.146 ± 0.001	0.14 ± 0.09	0.03 ± 0.07

We present some preliminary results from our *Exemplary Use Case*, the meteorological use case. Two datasets are onboarded: a CSV file containing stations information, and a PostgreSQL dataset containing historical data. We present the execution time of queries targeting only the CSV, only the PostgreSQL, and both in a join query. Average runtime across five runs with the standard deviation are presented in Table 1. Values marked with asterisk (*) represent query that did not get any results due the self join memory limitations. The evaluation has been made inside a Kubernetes development environment with 31 GB (allocatable) of RAM and 16 CPU cores. The Ontop pod was deployed without explicit CPU/memory resource limits. These experiments highlight the need of defined primary keys for the dataset to be queryable. Additionally, we can see that query executed on CSV are slower than on postgres, while the dataset is smaller, illustrating the need of using performant file-formats, such as Parquet. Besides query performance, we also evaluated the time required to automatically generate the ontology and mapping files from the input data sources. The mapping and ontology generation process took respectively 0.408s and 0.911s for the CSV and PostgreSQL datasets. This preprocessing step is executed only once during the onboarding phase and therefore does not affect query execution times.

3. Discussion, Limitations, and Future Work

We present an approach to generate an ontology and mappings to integrate data sources into a Virtual Knowledge Graph by exploiting automatically-generated metadata-annotations in the Croissant format. Croissant is a standard vocabulary for defining metadata for machine learning datasets. Preliminary results show that the Croissant profile can be used to generate an ontology and mappings for both CSV files and relational databases, as well as lenses file to annotate primary key information when provided. This work provides encouraging results for automatic data integration for scientific data in data lakes. At the same time, our preliminary investigation, although limited, already highlighted important performance bottlenecks to be addressed. First, the importance of being able to identify primary keys or candidate keys for each data source. Second, the necessity to convert CSV files to more performant file-formats (e.g., Parquet). Hence, in our future work, we will extend our approach to additional data types, such as JSON files, and plan to build a full-scale benchmark for the described setup. We also plan to explore the automatic discovery of primary and foreign keys in CSV files, by introducing a preprocessing step based on existing discovery tools [19, 20] or even LLMs. Finally, we will explore progressive ontology alignment and mapping processes for registered data sources.

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Declaration on Generative AI

During the preparation of this work, the authors used ChatGPT, and Grammarly in order to rephrase some of the sentences, fix grammar and spelling issues. After using these tools and services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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